

B4 Trajectories

Complete the values in the table below, assuming that all projectiles are launched horizontally and fall downwards with an acceleration of 9.81 m s^{-2} .

	Horizontal distance* /m	Horizontal speed / m s^{-1}	Time to target /s	Distance fallen /m
B4.1	25	75	(a)	(b)

* to target

- B4.2 You are trying to drop essential survival supplies from an aeroplane to help villagers isolated after an earthquake. You are flying 100 m above them, and your aircraft can travel no slower than 130 m s^{-1} . You fly on a straight line which will pass over the village. How far (in metres) in advance of over-flying the village do you need to drop the package?
- B4.3 A tennis player strikes a ball at a speed of 26 m s^{-1} at an angle of 8.0° above the horizontal from a height 1.20 m above the court. How far away from the player will it strike the ground? Assume that it doesn't hit the net. Hint: you may find it helpful to calculate the vertical component of the ball's velocity just before it hits the ground before attempting to calculate the time.

CHAPTER B. MECHANICS

Answers

B4.1

a) $\frac{25}{75} = 0.333 \text{ s}$

b) $\frac{1}{2} \times 9.81 \times 0.333^2 = 0.545 \text{ m}$

B4.2 $t^2 = \frac{2 \times 100}{9.81}$

$$x = 130 \times \sqrt{\frac{200}{9.81}} = 586.98 \text{ m}$$

B4.3 Final component of vertical velocity

$$v_y = -\sqrt{(26 \sin 8^\circ)^2 + 2 \times 9.81 \times 1.2} = 6.0529 \text{ m s}^{-1}$$

$$\text{Time } t = \frac{26 \sin 8^\circ + 6.0529}{9.81} = 0.9859 \text{ s}$$

$$\text{Horizontal distance } x = 26 \cos 8^\circ \times 0.9859 = 25.38 \text{ m}$$